JANNATUL FERDOUS MIM

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RESEARCH SUMMARY

Final-year CSE student (AI/ML major) and student researcher at North South University's TriModal AI Lab and ELITE Research Lab, working on trustworthy, human-centered AI in Bangla and English. My work rests on careful quantitative analysis: as co-first author of CultAct-ML (ORACLE @ EMNLP 2026), I ran the stratified ablation showing that a reasoning model given Bangla scenarios planned in English 70.1% of the time, a drift that a pooled analysis hides entirely, and tested each per-language shift with bootstrap confidence intervals. On PRECEPT (under review at NAACL 2027), I helped build and evaluate a multi-agent framework that makes medical LLMs answer reliably across rephrased questions in English and Bangla, and ran its agent ablation. As co-first author of MIMAL, I led a training-free detector of AI-generated scientific diagrams that reaches 85.5% accuracy where eight vision-language models fall below a constant guess. Champion of NASA Space Apps 2023 and 1st Place at the IBM Granite Hackathon.

Research Interests: trustworthy and human-centered AI, evaluation of LLMs and AI chatbots, multilingual and Bangla NLP, AI for health

EDUCATION

North South University — B.Sc. in Computer Science and Engineering | *Spring 2022 – Present*
Major in Artificial Intelligence and Machine Learning | Dhaka, Bangladesh

Udayan Higher Secondary School — Higher Secondary Certificate, Science | *2020*
GPA 5.00

Ahmed Bawany Academy School and College — Secondary School Certificate, Science | *2018*
GPA 5.00

RESEARCH EXPERIENCE

North South University — Student Research Assistant, TriModal AI Lab | *Nov 2025 – Present*
Dhaka, Bangladesh

CultAct-ML — cultural value defaults in multilingual LLM agents *(ORACLE @ EMNLP 2026; co-first author)*

- Co-first author of a plan-then-act benchmark that tests whether LLM agents default to Western values when they must commit to an action: 39 GLOBE-tagged value-conflict dilemmas in English, Indonesian, and Bangla, run on Qwen2.5-72B, Kimi K2, and Qwen3-235B for 1,053 valid trajectories. The Western-cluster default was significant in seven of nine model–language cells against a chance baseline of 0.468.
- Ran the paper's ablation study, stratifying plan drift by the language each plan was actually written in. Pooled, Qwen3-235B's Bangla plans looked almost identical to its English ones (LaBSE similarity 0.938), but that figure was carried by the 82 runs the model wrote in English; on the 35 plans genuinely written in Bangla, similarity fell to 0.838, the largest drift in the grid. The English-planning rate climbed from 0% to 9.4% to 70.1% across English, Indonesian, and Bangla, identifying planning-stage language abandonment as the mechanism.
- Extended the ablation with a 95% bootstrap over the 39 scenarios (B = 20,000) on the per-language shift in Western preference: the direction is model-dependent (Qwen2.5-72B +0.043 in Bangla, Qwen3-235B −0.085, Kimi K2 near zero), but no per-language shift is significant, so the paper reports direction rather than magnitude.

PRECEPT — memory-driven multi-agent reasoning for reliable medical QA *(NAACL 2027, under review; co-author)*

- Helped develop PRECEPT, a four-agent framework that answers a medical question, reflects on each graded error, routes the resulting insight to a medical sector, and curates it into an episodic memory that conditions later attempts, improving reliability at inference time without fine-tuning. It raises ReAcc, ReCon, and accuracy for every model it is applied to, on ReMedQA in English and on the new item-aligned ReMedQA-Bangla.
- Ran part of the evaluation across the paper's eleven-LLM grid, from small general and medical models (≤8B) to large open and proprietary ones, each scored on seven meaning-preserving variants of every question.
- Ran the cumulative ablation that adds one agent at a time (GPT-OSS-20B, 100 MMLU College Medicine questions): reflecting on the current error and retrying raised ReAcc from 55.0% to 72.0%, each later agent let the retries fix more first-attempt errors (57 to 73), and the full system reached the highest accuracy (91.1%) while the curator agent shrank the memory by 62% (521 to 198 rules).

Applied systems

- Co-developed MediTalk AI, a multimodal medical pre-diagnosis assistant: built the intake pipeline that combines OCR for medical reports, speech recognition for patient consultations, and image analysis for scans, and defined a structured summary format that keeps output consistent across input types, so clinicians can review a case quickly while patient information stays in one controlled record.
- Co-developed Ask Alvi, a multi-model AI search engine on Next.js and Express that routes queries across multiple LLMs with retrieval augmentation and function calling, extended to image, music, and video retrieval and to chat-platform interfaces.

ELITE Research Lab — Student Researcher | *Sep 2026 – Present*
Remote

MIMAL — training-free detection of AI-generated scientific diagrams *(co-first author; led implementation and experiments)*

- Co-first author of MIMAL, a detector that needs no training: it splits a diagram into crops, drops blank ones, perturbs each along its Daubechies (db2) wavelet high-frequency band, and scores how the patch-token embeddings of a frozen DINOv2 respond, so a single unstable region can move the score. MIMAL reaches 85.5% accuracy and 0.851 AUROC, while eight vision-language models, including GPT-4o and Gemini 1.5 Pro, score 18.3–34.2%, below a 50% constant guess.
- Built SciFigBench, a balanced 1,220-image benchmark that pairs 610 real methodology diagrams from NeurIPS 2025 papers with counterparts generated by the agentic PaperBanana pipeline from the same paper context, so a detector must respond to how a figure was made rather than to what it shows.
- Ran the stage-wise ablation: replacing the global [CLS] embedding with dense patch tokens lifted accuracy from 55.5% to 71.3% and accounts for 82.7% of the AUROC gain, and the db2 wavelet with background masking added a further 14.2 points. Showed that the signal runs opposite to the one reported for photographs (synthetic diagrams are the stable ones), so photo detectors cannot be reused unchanged.

PUBLICATIONS

- *CultAct-ML: Measuring Cultural Value Defaults in Multilingual LLM Agents Under Action.* Ahmad Fahmid*, Jannatul Ferdous Mim*, Md Adnan Arefeen. ORACLE Workshop @ EMNLP 2026 (accepted). (*equal contribution) [code]

UNDER REVIEW AND MANUSCRIPTS

- *PRECEPT: Memory-Driven Multi-Agent Reasoning for Reliable Medical Question Answering in English and Bangla.* Under review at NAACL 2027. Introduces ReMedQA-Bangla, an item-aligned Bangla version of MedQA, MedMCQA, and medical MMLU.
- *MIMAL: A Self-Supervised Model Is a Training-Free Detector of AI-Generated Scientific Diagrams.* Ahmad Fahmid*, Jannatul Ferdous Mim*, Md Kishor Morol. ELITE Research Lab, 2026 (manuscript). (*equal contribution) [code]

SKILLS

Research Methods & Analysis: quantitative analysis and statistical testing (bootstrap confidence intervals, significance testing against chance baselines, stratified and ablation analysis), data cleaning and validation, evaluation protocol and metric design, benchmark construction, robustness evaluation of LLMs under question rephrasing, LLM-as-judge error analysis, Bangla–English multilingual evaluation, literature review, reproducible and documented pipelines

Programming & Data: Python, SQL, C++, C, Java, JavaScript, TypeScript; Scikit-learn; MySQL, MongoDB; FastAPI, Django, React.js, Next.js, Node.js; Git and GitHub, Linux, LaTeX

AI & Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, LoRA / PEFT, RAG, LangChain, sentence-transformers, LaBSE, multi-agent orchestration, agent memory and retrieval, vision-language models, self-supervised vision models (DINOv2), computer vision, OCR, speech recognition, deep learning for medical imaging

Languages: Bangla (native), English (professional working proficiency)

SELECTED PROJECTS

- **Accent Recognition System.** Fine-tuned Qwen models on accent-specific speech patterns, served inference through Groq and Mixtral for low-latency classification, and benchmarked against standard pre-trained models, iterating until the system beat those baselines on classification accuracy.
- **Research Paper Automation.** Retrieval-augmented pipeline (LangChain, OpenAI) that ingests research papers, answers questions against them, and returns sectioned summaries grounded in the source text, cutting the manual reading load in literature review.

AWARDS

- Champion, NASA Space Apps Challenge 2023.
- 1st Runner-Up, NASA Space Apps Challenge 2025.
- 1st Place, IBM Granite Hackathon.
- Top-30 finish in five further events: UIU Project Showcase, BUBT Hackathon, Solvio Hackathon, Lovable AI Hackathon, and Co-Creating with GPT-5 Hackathon.

LEADERSHIP AND CO-CURRICULAR ACTIVITIES

NSU Computer and Engineering Club — General Member, Corporate Team | *Jan 2022 – Present*

- Run the corporate team's communication with sponsors and guests, and host corporate guests at flagship club events, owning arrangements from arrival to close.
- Organise professional development sessions with core members, coordinating scheduling and on-day logistics end to end.

NSU ACM Student Chapter — General Member, Logistics and Team Provision | *Jun 2022 – Feb 2023*

- Handled procurement, distribution, and asset tracking for cross-functional teams at major events, so requirements were covered ahead of time rather than patched on the day.

REFERENCES

Dr. Md Adnan Arefeen, Assistant Professor, North South University. Email: adnan.arefeen@northsouth.edu

Dr. Sumaiya Tabassum Nimi, Assistant Professor, North South University. Email: sumaiya.nimi@northsouth.edu